

Name: _____

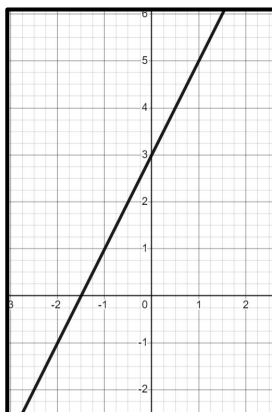


Finding the equation of the line $y = mx + c$

Task A: Writing the equation of a line from key features

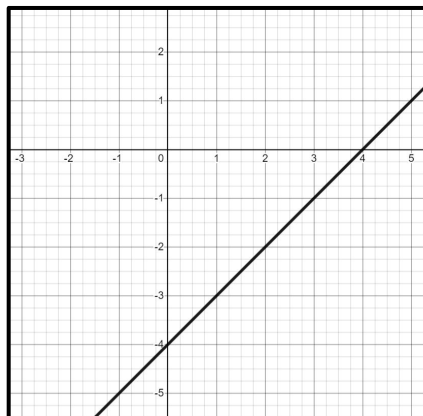
1) For each graph, work out an equation for the line in the form $y = mx + c$

a)



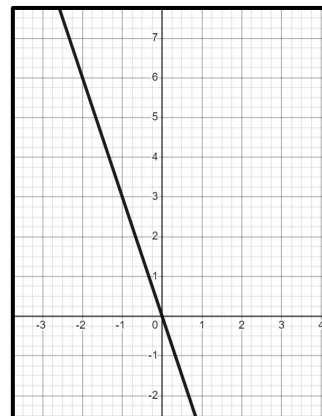
gradient:
y-intercept:
equation:

b)



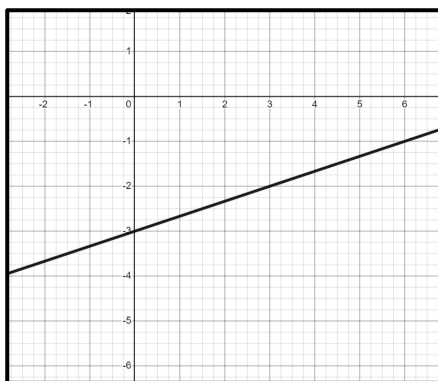
gradient:
y-intercept:
equation:

c)



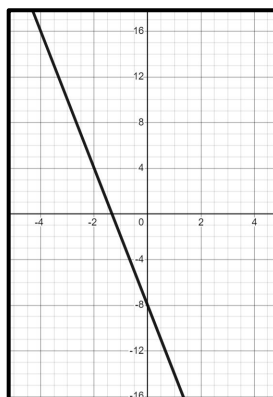
gradient:
y-intercept:
equation:

d)



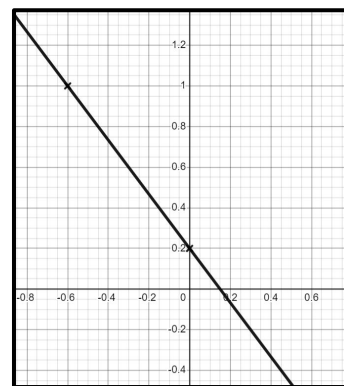
gradient:
y-intercept:
equation: $y =$

e)



gradient:
y-intercept:
equation:

f)



gradient:
y-intercept:
equation:



Task B: Finding the equation of a line

1) Calculate the equation of the line which has the table of values shown.

a)

x	0	1	2
y	-7	-3	1

b)

x	-1	0	1
y	14	3	-8

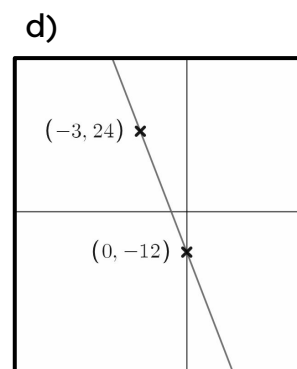
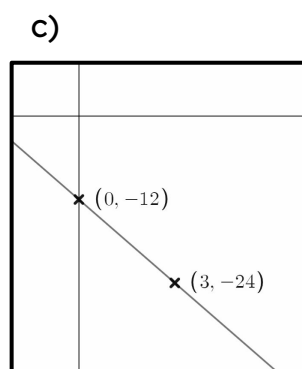
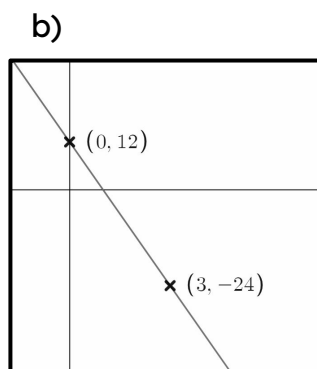
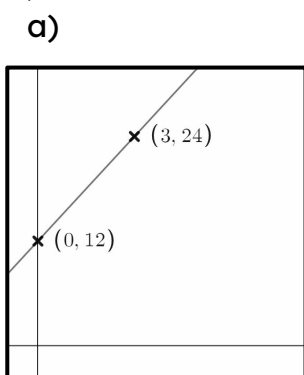
c)

x	...	1	2	3
y	...	5	3	1

d)

x	...	10	11	12
y	...	-12	-11	-10

2) Calculate the equation of each line.



3) Calculate the equation of the line passing through the coordinates given

a) (3, 5) and (5, 13) b) (3, 5) and (7, -15) c) (3, 5) and (-2, 10) d) (3, 5) and (-15, -1)

Change x	Change y

Change x	Change y

Change x	Change y

x	0 ...	3 ...	5
y		5 ...	13

x	0 ...	3 ...	7
y		5 ...	-9

x			
y			

Equation:

Equation:

Equation:

Equation: